

	HINDUSTAN SHIPYARD LIMITED (A Govt. of India Undertaking) Ministry of Defence Gandhigram, Visakhapatnam - 530005	Electrical Design	REV: 002
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SHIP BUILDER
HINDUSTAN SHIPYARD LIMITED
 VISAKHAPATNAM- 530005.

**PURCHASE ORDER TECHNICAL SPECIFICATION FOR SOUND POWERED
 TELEPHONE SYSTEM**

Supplier: M/s Elcome Integrated Systems, Navi Mumbai

Project	:	DIVING SUPPORT VESSEL VC 11190 – 91
CLIENT	:	IHQ MOD (N) / DSP
DOCUMENT No	:	HSL/DSV/EL/SOTR/009

REV	DESCRIPTION	DATE	Prepared By:	AUTHORISED BY
002	Updated as POTS	15.04.2021	R.Sreenu	B. Ravi Kumar
001	Incorporated IHQ, MoD (N) comments, Letter Dtd. 13 Aug 20	26.08.2020	R.Sreenu	B. Ravi Kumar
000	First Issue to IHQ, MoD (N)	27.06.2020	R.Sreenu	B. Ravi Kumar

ANNEXURE-1	Scope of Supply for Sound Power Telephone system	
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Chapter 1 - Preamble	
Clause No.	Description of Technical Specification
1.1	This document defines the technical requirements for design, development, Manufacture, supply, testing, installation, setting to work, trials and onboard commissioning of sound powered telephones complete with associated accessories and fitted on suitable mounts for 2Nos. of Diving Support Vessels (DSV) class of ships.
1.3	System Overview: The SP Telephone to be designed to offer an easy to use, low maintenance communications facility, ideally suited to emergency or secondary communication applications. The SP telephones should be technologically upgraded one to realize quality audio with pre amplifier working on chargeable capacitor battery. The capacitor battery charged by the call generator (by cranking) provides power to the integral amplifier for up to 20 minutes of operation. The hand operated generator to initiate a low frequency pulse DC calling signal, which is converted at the receiving telephone into ringing tone. The SP Telephone systems should have multi-way selection facility amongst users within a functional line.
1.4	System Feature: SP Telephone, an emergency communication set will be catered for various positions as detailed in the groups. Those SPTs, being multiway type any position can communicate with all other positions within a functional line. The SP Telephone system will have the following major features (a) SPT system should provide loud and clear voice communication between key locations on a ship for both emergency and routine communication. Excellent speech quality to be achieved by integrated pre-amplification circuits & microphone with dynamic noise canceling and negligible power loss over cabling. (b) SPT should be light, robust and corrosion resistant and will be suitable for all on-board naval environment. (c) SPT system should operate independently without ships/ external power supply and provide seamless voice communication during emergency conditions. SPT will have a provision for connecting the ships emergency DC supply (24V) and should be able to operate as conventional external powered telephone system, thereby avoiding the use of hand cranking and to work as a battery telephone system. (e) SPT system should have internal systems/ circuitry to amplify the voice signals and able to transmit/receive voice signals between the other SPTs installed onboard. (f) The SPTs control box is to be equipped with visual (LED) and audible call indication. The audible call and LED indication should be prominent till a distance of at least 10 m from the unit during ship's operating phases. Suitable indications to be provided to identify caller and select a particular user for calling is required to be provided in the SPT control box. (g) SPT system should have noise compensated headsets with noise compensated microphones that can be connected to a plug/socket box fitted

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	in the SPT system network. Headset directly to be connected to the telephone station or connected to a portable Telephone via plug box. (h) The SPT system should have plug/socket box for connecting the portable SPTs and portable headsets. The location and number of plug / socket box required for connecting the portable SPT / headsets. (i) The SPT system should have audio and visual combined flash alarm units. These units will be connected to the individual SPTs. The requirements of audio and visual combined flash alarm units in the SPT units will be interfaced with the individual SPT directly or through a relay box. (j) The SPT system has number of variants with different configurations Viz single way, multiway. The SPT system should provide the means to select one of the various positions with which the user may need to communicate. (k) The SPT system should have group isolation (Anchoring group, Conning and Steering group, Electrical & communication group, Machinery group, Helo & AVCAT group, Diving & DP group and Damage Control Group) and one group should be able to communicate with the SPTs of other groups. Call Conference facilities maximum upto 08 SPT should be available. (l) The SPT system for new constructions ships should have screened cable, network for interconnection between SPTs. (m) The SPT and associated accessories should meet the ingress protection based on locations where it is going to be fitted. For SPTs fitted in the weather Decks should have IP66 rating for underwater compartments it should have IP67 rating and for other locations it should have IP55 ratings. The enclosure should be an aluminum or steel or alloy or composite materials having suitable arrangements for deck head/ Bulkhead mounting type installation on-board ships. (n) If one SPT is sited inside a hazardous area, the SPTs shall meet the degree of intrinsic safety required for hazardous areas, thus preventing dangerous voltages being introduced into the hazardous area by telephones/ Circuits of the SPT system network sited outside the area.
	Chapter 2 - Functional Requirements
2.1	The Supplied SPT shall have the following major features: - (a) SPTs shall provide load and clear voice communication between key locations on the ship for both emergency and routine communication. Excellent speech quality shall be achieved by integrated pre-amplification circuits and negligible power loss over cabling. (b) The SP Telephones are Light & Robust and to be housed in a corrosion and impact resistant casing which can be desk-top, bulkhead or with an optional mounting kit for console mounting. (c) SPTs shall be multiway type with selection switch for selecting users (i.e. flexibility in operation). Any user shall be able to communicate with all other users within a functional line. (d) SPT system shall be able to operate independently without ship's/external power supply and provide seamless voice communication during emergency

	conditions. SPTs shall have a provision for connection to ship's DC supply (24V) and operate as a conventional external powered telephone system, thereby avoiding use of hand cranking. (e) SPTs shall be equipped with visual (LED) and audible call indication. The audible call and LED Indication shall be prominent up to 10M distance from the unit during ship's operating phases. Suitable indications shall be provided for caller identification and selecting a particular user for calling. (f) SPTs shall have noise compensated headsets and noise compensated microphones connected directly to the SPT Stations. (g) SPT system shall have group isolation between following groups as per table below:- <table><tr><th>Group Name</th><th>Functional Description</th><th>No. of User Position</th><th>JB Requirement</th></tr><tr><td>A</td><td>Anchoring, towing & docking</td><td>3</td><td>Dedicated JB</td></tr><tr><td>C</td><td>Conning and Steering</td><td>9</td><td>Dedicated JB</td></tr><tr><td>E</td><td>Electrical & Communication</td><td>17</td><td>Dedicated JB</td></tr><tr><td>M</td><td>Machinery</td><td>11</td><td>Dedicated JB</td></tr><tr><td>H</td><td>Helo & AVCAT</td><td>7</td><td>Dedicated JB</td></tr><tr><td>DI</td><td>Diving & DP</td><td>15</td><td>Dedicated JB</td></tr><tr><td>D</td><td>Damage Control</td><td>8</td><td>Dedicated JB</td></tr></table>	Group Name	Functional Description	No. of User Position	JB Requirement	A	Anchoring, towing & docking	3	Dedicated JB	C	Conning and Steering	9	Dedicated JB	E	Electrical & Communication	17	Dedicated JB	M	Machinery	11	Dedicated JB	H	Helo & AVCAT	7	Dedicated JB	DI	Diving & DP	15	Dedicated JB	D	Damage Control	8	Dedicated JB
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2.2	Supplied system shall meet the following major specification:- <table><tr><td>(a)</td><td>Construction</td><td>Enclosure shall provide high impact resistance, stability to ultra violet light and excellent chemical resistance.</td></tr><tr><td>(b)</td><td>Calling Signal Level</td><td>>75 dBA at 1 Meter or better</td></tr><tr><td>(c)</td><td>Switch</td><td>Magnetically operated reed switch</td></tr><tr><td>(d)</td><td>Audio level</td><td>-13 dBm or better</td></tr><tr><td></td><td>Mechanical requirements</td><td>Dimensional of SPT sets & JB's shall be kept as minimum as possible.</td></tr></table>	(a)	Construction	Enclosure shall provide high impact resistance, stability to ultra violet light and excellent chemical resistance.	(b)	Calling Signal Level	>75 dBA at 1 Meter or better	(c)	Switch	Magnetically operated reed switch	(d)	Audio level	-13 dBm or better		Mechanical requirements	Dimensional of SPT sets & JB's shall be kept as minimum as possible.																	
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	Chapter 3 - System Configuration																																
3.1	Muliti-way SP Telephone: SPT shall be COTS type suitable for marine applications. Muliti-way SPTs of 12,24 extensions shall be supplied along with noise cancelling handset. It shall be provided means to select one of the various positions with which the user may need to communicate in a single functional line through selector switch.																																

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3.2	Junction Box: COTS type Junction Boxes suitable for marine use shall be used for interfacing the SP Telephones within a functional line. JB terminals, dimensions, connection for a single functional line within the JB etc. shall be indicated in the drawing.
3.3	Headset: The headset with integral noise cancelling microphone shall be used in acoustically harsh environments or where hands-free operation is required (i.e. Noisy areas and weather deck). Stowage clips are to be provided near the SP Telephone sets.
3.4	Flasher Hooter and relay Box: SP Telephones shall have provision for suitable audio-visual indications to annunciate incoming call at noisy locations by means of an audible bell and a visual indication. Relay box shall be used to connect the flasher hooter to the SP telephone.
3.5	Directory Plates: Directory plates must be provided alongside each telephone unit. Diagram plate indicating connection details shall be affixed on the equipment directory shall be in aluminium anodised plate.
	Chapter 4 - Scope of Supply
4.1	As mentioned the system functionality, the types of groups and the number of positions for each group, firm shall supply the required units and subunits to ensure the system functionality. The scope of supply is placed at Annexure-1.
	Chapter 5 - Technical Requirement
5.1	Power supply: Ship's power supply of 230V AC, 1Ph, 50Hz shall be made available to the relay box flasher hooter unit. The SPT system shall have a provision for connecting to ship's 24V DC supply, thus operating as a conventional telephone system without the use of hand cranking.
5.2	Internal wiring: Internal wiring of the equipment shall use LHF wires.
5.3	Cable: System Cables are excluded from firm's scope of supply. Any other special cables required shall be supplied by the firm. Shipyard will indicate Cable part no and OD post placement of order. Inter unit cable diagram (Line diagram) specifying type of cable used to be provided by the firm along with the Binding Data.
5.4	Cable Entry Glands: Cable entry glands shall conform to specification BS6121/ EN 624440-2013 for incoming and outgoing cables. Cable glands shall be fitted with the equipment, the gland size shall be suitable to the OD of the cable indicated by the Shipyard.
5.5	Terminal: Bolted type/WAGO type terminal & crimped socket of electrolytic copper shall be provided for all incoming and outgoing cables. Adequate spare terminal strip shall be provided for provision of additional control and indication facilities. Adequate space shall be provided inside the equipment for bending and termination of incoming & outgoing cables.
5.6	Earth Terminals: An external earth terminal of brass or stainless steel is to be provided for devices and control panel.

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5.7	Installation Material: One set of installation material per ship inclusive of special tools if any, special cables if any, plugs, adapters, mating flanges, sockets (with accessories), pins and cable lugs, vibration mounts etc. as required for satisfactory completion of system installation on board ship shall be supplied.
5.8	Components: All components shall be of Naval (Marine) approved type. Standard specification and grade of material of each component used shall be indicated in the binding drawings. Standard specification and grade of material of each component used shall be indicated in the binding drawings. Make and type of components are to be listed in the binding drawings. Indication lamps used shall be of LED type. The equipment shall be designed so that all components may be removed from the front of the cubicles with the front cover only removed. There shall be no hidden connection. All the components of the equipment shall have easy access for removal/ refit.
5.9	Enclosure: The enclosure shall be aluminium/steel/alloy/composite materials having suitable arrangement for flush/bulkhead mounting type installation on-board the ship. Subunits shall conform to Ingress Protection IEC 60529 viz. IP22 for indoor units, IP44 for machinery compartments and IP56 for outdoor units.
5.10	Lifting Arrangement: Equipment weighing more than 40 Kg shall be provided with collar eyebolts or suitable lifting lugs/arrangements. If the eyebolts cannot remain in site after the equipment has been installed in the ship, provision shall be made for their securing arrangement on the equipment.
5.11	Painting: Enclosure finish colour shall be admiralty grey, semi-gloss as per approved painting specifications/procedures. Painting shall conform DGS 251/RAL 7001
5.12	Talley and diagram Plates: Tallies and diagram plates for all equipment shall be made of anodized aluminium alloy/SS. Size of tally plate and their letters shall conform to specification NES-723. 230V danger tally shall be provided on the equipment at a permanent place where required. Safety markings on main equipment and associated devices/units shall conform to NES 784.
5.13	EMI/EMC Standard: The equipment shall be capable of operating in the vicinity of other equipment / systems without causing or responding to undesirable electro-magnetic energy. The equipment design shall consider electromagnetic interference (EMI) coming from the external sources, generated from within its own enclosure and degrading the performance of the equipment. EMI may be reduced by judicious selection of components and use of good engineering practices like grounding, bonding and shielding. a) COTS equipment shall comply IEC 60945/IEC 60533 or equivalent EM standards. b) All equipment shall have the provision of grounding, bonding and shielding as per NECP 500 and MIL-STD-1310H.
5.14	Permissible Temperature Rise: The equipment must be designed so that the maximum permissible operating temperature of components is not exceeded during normal operating conditions in a maximum ambient temperature as specified in this specification.
5.15	Cooling: Experience in onboard use indicates a significant proportion of static conversion equipment failures occur as a result of overheating. Sufficient cooling arrangements by natural air convection shall be provided in the equipment in order to meet the permissible limit of temperature rise.

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5.16	Reliability and Maintainability (R&M): (a) Mean Time Between failure (MTBF): The design of the equipment shall be such that the estimated mean time internal between failures shall be greater than 10000 hours. (b) Self Noise & vibration: Care is to be taken in the design and manufacture to ensure that the air borne noise and structure borne vibrations produced by the units/ devices, the self-induced air borne noise and vibration level shall meet ships stealth features. (c) Mean time to Repair (MTTR): System shall be designed for low MTTR not exceeding 15min.
5.17	Control and Protection: The following protection principle shall apply to all electrical and electronic equipment: a) Internal fault protection is to: (i) Prevent further damage within the equipment, subsequent to a fault (ii) Prevent damage to the converted power loads (iii) Prevent the primary supply going out of specification. b) External fault protection is to: (i) Prevent damage to the equipment, subsequent to a fault external to the equipment. (ii) Where Practicable, include current limiting techniques to allow the equipment to work with a short circuit onto the output. (iii) Be defined in terms of conditions under which the equipment shall trip. c) Estimated and calculation of the effects of fault conditions are to include: (i) Stored energy in the load (e.g. resulting from shunt capacitors or motor loads) (ii) Stored energy in the equipment itself (e.g. in Back-up Batteries) d) The design of protective arrangements shall take into account thermal capacity of the components being protected and heat sink arrangements incorporated within equipment. The contribution to R&M figures of protective arrangements must be considered.
5.18	ESD Protection: The system design shall take into account adequate measures for Electro Static Discharge (ESD) control and protection at PCB/ module/ assembly and unit level. Each Electro Static Discharge sensitive part/assembly shall be duly marked with a symbol/ warning. The firm shall use ESD protective materials for handling, packaging storage and transport.
5.19	Environmental Stress Screening (ESS) Test: ESS test for electronic components/sub-systems is required to be carried out for 100% components/ parts/ assemblies as per IHQMOD(N) Policy letter 66301/Policy-07/DQA(N)/QA-07 dt 09 Aug 16
5.20	The offered SPT system shall comply with EED-50-08 (R2). Alternatively the SPT shall be supplied with Class type approval certificate confirming to IEC Standards.
5.21	Shock requirement of the SPTs is to comply with IEC 60068-2-27 / or equivalent

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	standard (severity applicable for merchant ships) iaw clause 6(f) of EED-50-08 (R2).
5.22	Humidity and heat run tests to comply with IEC 61347-1.
	Chapter 6 – General
6.1	Ship's Data: The System shall be suitable for fitment on DSV class of ship with the following features:- a) Length Overall - 119.40m b) Length Between Perpendiculars – 110.70m c) Breath MLD – 22.80m d) Depth to Deck10 – 10.40m e) Design Draft – 6.00m f) Total compliment - 215
6.2	Classification Society: Vessel Shall be designed, built and outfitted as per applicable provisions and regulations as given in the Naval Rules of Indian Register of Shipping. The Class Notation for the Diving Support Vessel as follows. Swastika SUL "Diving Support Vessel", FDA, VIB(HGL), HELIDK, RAS(AB) Swastika IY, DP(2), EP, DSV Descriptive Note: "Submarine Rescue vessel", "Deck strengthened for 5t/m2", Special Government Service".
6.3	Operating Condition: The system devices/equipment shall operate satisfactory in the marine environmental (coastal conditions) and other operating conditions enumerated in the succeeding paragraphs.
6.3.1	Seaway conditions: a. Roll : Max 20° with 10 sec period. b. Trim : Max 5° c. List : Max 20° from vertical. - Pitch : Max 6° with 20 sec period.
6.3.2	Sea Worthiness: Main equipment along with its control equipment & accessories should meet sea worthiness criteria as mentioned below. The sea worthiness of the ship shall be as follows:- a. Sea worthy for transit on all headings upto Sea State 7 and survivable on best headings at Sea State 8. b. Capable of operating Helicopter in Sea State 6 on best heading. c. Capable of conducting Diving Operations upto Sea State 4. d. Capable of Dynamic Positioning operations upto Sea State 4.
6.3.3	Units and Language: Language shall be English. Metric Units shall be used.

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6.3.4	Environment Condition: The equipment and the accessories including all integral and associated service system should be capable of continuous operation when the ship is in the following conditions: - Ambient Conditions : - Sea water temp : 1 deg.C to + 35 deg.C - Max relative humidity : 100% at 38 deg C - Salinity of water : upto 35 ppt Max. Machinery room temp: Machinery room compartments maintained at ambient (33°C) + 5 in open condition and between 45 to 55°C in closed condition.
6.4	Commonality: The firm to mention the level of commonality achieved within the system at the following levels :- - Module/Sub- module level - PCB level - Component level
6.5	Weight Data Sheet: The firm to furnish Weight data sheet for Equipment detailing the following for the use of calculation for weight curve :- - Carcass weight - Total weight of equipment - Centre of gravity showing 3 axis coordinate
6.6	Security: The information about the ship and the requirements mentioned in the SOTR are not to be communicated to any third party without prior approval of IHQMOD(N).
6.7	Warranty: The Equipment with Associated controls/ instrumentation shall be guaranteed for stipulated performance for a time period mentioned as per STAC/Commercial Clause. The equipment supplied shall be warranted from defects arising due to the manufacturer/firm and performance for the said period and cover all the defects arising from malfunction through design faults, inappropriate material, bad production and non-conformance to specifications. Any expense on account of repair/ supply of spares against guarantee defects shall be borne by the firm.
6.7.1	Extension of warranty: Firm shall mandatorily indicate in the offer, the annual rate for extended warranty period, in the event of expiry of warranty period as detailed above along with the associated commercial terms and conditions if any.
6.8	Upgrades and Obsolescence / Product Support: Latest version of the equipment/ system at the time of tendering shall be provided. All upgrades and modifications thereafter in the hardware and software of the system need to be communicated to the Navy for the complete support life of 25 years from the date of commissioning of the equipment on board ship excluding electronic item for which the period is 10 years. Obsolescence of subsystem, components, parts and assemblies need to be communicated to Navy at least two year in advance with recommended remedial action. A maintenance contract may be offered for consideration and negotiation at the later date. Firms supplying imported equipment/systems are to ensure that product support and aftersales service in India are available for all assemblies/ modules/ software including

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	third party supplies prior commencement of trials.
6.9	Safety standards: Equipment shall offer total safety to personnel from moving parts and other associated equipment from vibrations. Safety standards as per MIL STO 454/ equivalent shall be met.
	Chapter 7 - Spares, Tools and Test Equipment
7.1	Onboard Spares : OEM recommended onboard spares for schedule servicing, maintenance & break down maintenance falling due within the first operation cycle for 02 years shall be provided for the equipment in ILMS Ver 2 format (Integrated Logistics Management System) indicating the item code number, quantity and item wise cost. The Spares consumed within the guarantee period to rectify / liquidated known defects shall be replaced free of cost. The supplier shall prepare the list of OBS (including onboard maintenance tools) based on the Reliability and Maintainability data and taking into account IN's maintenance philosophy, and shall forward to IHQ MOD (Navy) for vetting. The complete inventory of system parts including OBS & B&D spares is to be provided in INCAT (Indian Naval Catalogue of Inventory) compatible format in electronic media for ILMS (Integrated Logistics Management system) of Indian Navy, for the management of spares. The supplier shall provide both hard copy as well as soft copy.
7.2	Five Year Base and Depot (B&D) Spares. Base spares recommendation shall cover maintenance / overhaul requirements of 5 years including two refits shall be provided for the equipment in ILMS Ver 2 format. Recommendations for insurance holding of long lead items are also to be indicated. Itemized cost of B&D spares are to be forwarded with the main offer along with part identification list of spares. The list would contain price, description, Pattern no. and quantity fitted on each equipment in respect of various parts / components. The quotation of spares shall be valid for at least 18 months. The Buyer shall subsequently range and scale the B&D spares. The cost negotiation of B&D spares would be carried out by a committee constituted by the Buyer, wherein the seller's and sub-vendor's representative would be a member. The OEM shall undertake procurement of the ranged and scaled spares as per the ordering instructions issued by the IHQ MoD (N).
7.3	Installation & Commissioning Spares: The supplier's service of engineer shall supply/ carry one set of installation and commissioning spares along with the main equipment.
	Chapter 8 - Binding Data , Design Drawing & Technical Documentation
8.1	Binding Drawing: The firm shall submit approved and as made drawings sufficiently detailed to show the manner of construction and operation, the method of assembling and dismantling. The following binding drawing/documents in duplicate (both hardcopy & softcopy) shall be submitted to IRS for approval with a copy to shipyard within 4-6 weeks after placement of the order. (a) Installation drawings indicating overall dimensions, C.G., weight, maintenance

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	envelope, IP rating, power ratings, vibration mount details etc of each unit. Detailed foundation drawing including bolting plan shall be included. (b) Mounting arrangement drawings with installation instructions. (c) Heat Dissipation of individual units in compartment & in close loop ventilation system, as required for the system. Other interface requirement from ship viz. air, water, cooling etc to be indicated. (d) Material list Indicating Part No., Quantity, Material, Makers name etc. (e) Block diagram of all major assemblies showing interconnection between these assemblies as well as cables specifications. (f) Power supply scheme for the system with specific mention of UPS requirement. (g) Cable length limitation etc, if any. (h) Recommended arrangement of devices in nominated compartments. (i) Circuit diagram, wiring diagrams, wiring schedules, terminal connection and core connection details. (j) Details of standard and optional factory tests
8.2	Design Documents/Drawings: All associated design documentation, drawings and equipment list applicable must be in English language. All drawings and documents shall contain dimensions and other parameters in metric units (SI Units). The drawing and documentation shall be complete to enable installation on board. These shall cover all subassemblies and accessories of the equipment. The documents/drawings shall be provided in both hard copy form and on CD_ROM. Following draft documents shall be presented for approval of IHQMOD(N): a) Technical and operation Instruction Manual b) Onboard Maintenance Manual c) field and Depot Maintenance manual d) Operation Maintenance document e) Installation and testing specifications/manual f) Certified Test Data and report g) Parts and Tools Catalogue h) Installation drawings in reduced Size i) As Fitted Drawings j) Any Other Special Instructions for Installation and Preservation k) Conformation of Standards Specified l) Test Procedures and Documentation m) Heat Dissipation of Devices n) Storage, handling, transportation details etc. o) Details upto 2nd level maintenance for repair and troubleshooting. Component level maintenance details to be provided along with list of components. p) The software in the system shall comply with any of the established standards like MIL-498, IEEE 12207 etc. All Documents as per the standards related to software design, setting and maintenance shall be provided.

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8.3	Technical Documentation: Details of technical documentations to be supplied are as mentioned below: Separate manuals in respect of following for both equipment and system shall be provided in hard copy:- (a) Operation and Specifications including onboard Repair and Maintenance. This should include information about the equipment (such as serial number, trial reports, etc) and the manufacturer (such as addresses, telephone and fax numbers, email, etc). (b) Repair Technical Document with detailed drawings (c) Complete Parts List		
	S.No	Description	No. of Copies
	1	User Handbook (Including Technical Description, operation Instruction and Service log Book)	Two sets of documents shall be supplied in CD_ROM in addition to nine sets of hard copies for each ship.
	2	Technical Manual including Setting to work & test Procedures.	
	3	CPIL/PIL	
	4	Servicing and Maintenance Manual (field and depot)	
	5	Unpacking and Installation specifications/ drawings	
	6	Recommended list of OB spares, testing tuning spares, tools and test jigs.	
	7	Installation checks, HAT/SAT schedules	
	8	As made drawings including winding & performance data	
	9	Test certificates	
	Chapter 9 - Quality Assurance		
9.1	The detail design, material and workmanship shall be in accordance with the best worldwide recognized marine practices, to ensure reliability, durability and ease of maintenance which comply with the ship's requirements. The design shall be such that weight and size are reduced to the minimum practicable, ensuring no compromise in reliability or significant design criteria. A Quality Assurance Program shall be specified by the firm in this offer.		
9.2	Quality Assurance Program:		
9.2.1	Details of test equipment, test methods, preliminary qualification tests, etc shall be indicated. All tests shall be carried out by firm at his own works, in presence of and inspecting authority, prior to dispatching the equipment to the customer.		

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9.2.2	Quality assurance plan proposed and any subsequent amendment shall be discussed and agreed upon between IRS & the firm prior to placement of order.
	Chapter 10 - Inspection & Testing
10.1	Inspection: (a) For Design : Directorate of Naval Design (SSG) / Directorate of Electrical Equipment (DEE) of Indian Navy (b) For Installation : Warship Overseeing Team (WOT) of IN at HSL, Vizag (c) For ship trials : IRS (d) For QC : IRS and IHQ MoD(N) Rep (e) For approval of HATs/SATs schedule: IRS
10.1.1	The product offered by the firm shall conform to standard engineering practices. The equipment shall be subject to stage inspection and final test and trials by IRS. Any deviation from the mentioned specifications shall have to be brought to the notice of IHQMOD(N).
10.2	FATs: FAT for the first system shall be carried out at OEM premises by the reps of classification society including QA representative against ATP mutually agreed by IRS and OEM. During FATs firm would demonstrate system shall meet the functional and performance requirements indicated as per approved SOTR. Firm shall submit FAT document to IRS within 03 months after placement of order for approval. The FATs procedure shall comprise of: (a) Functional test (b) Verification of design, especially firmware and software (c) Test equipment used, calibration requirements. (d) Pass / Fail Criteria. (e) Expected duration and time phasing
10.2.1	One copy of FAT report shall be submitted to IHQMOD(N) and shipyard for record/ comments.
10.3	HATs/SATs: HAT/SAT Schedule shall be prepared by firm and forwarded to IRS for Approval within 06 months of Purchase order issued to firm.
10.4	Service of Engineers: The firm shall provide the necessary service of engineer to carry out the following activities: a) Setting to work b) Harbor and Sea Trials
10.4.1	Firm shall depute firm's experienced, qualified and expert engineer/technician to carry out the above activities onboard in-line with ship's construction timeline for the system. Above activities shall be undertaken to ensure satisfactory performance of the system onboard and for proving the system to the inspection agencies as per approved test procedure.
10.5	System Performance responsibility: The firm shall present the system for performance Evaluation Trials (HAT/SAT) to prove the expected performance assured. In case of any irregularities in the operation/ performance of the system or non-conformance to

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	specified parameters observed on integration with ships system, the firm is bound to rectify the defect. The firm shall ensure complete responsibility of satisfactory operation of the system on board.
	Chapter 11 - Other terms and Conditions
11.1	Shipping: All equipment shall be adequately packed and protected with supports to ensure adequate protection during all modes of transportation. Each unit within a package/container shall be clearly marked for identification. The container shall clearly indicate the item description with caution marks, quantity, weight, size, etc. A separate document giving complete details and instructions for storage, preservation, handling and transportation after delivery shall be supplied. The firm shall indicate the delivery schedule, port of embarkation, transport, packing, preservation, insurance, etc.
11.2	Training: The OEM shall conduct an operational and maintenance training course at the shipyard premises to the ship's crew / maintainers. Complete training package for on board exploitation, maintenance and for second level maintenance along with tools and test kit to be offered. The level and number of personnel recommended for training will be indicated. All training material shall thereafter be handed over to Navy for future training requirements. The training schedule shall be prepared in consultation with WOT/Ship Staff well in advance.
11.3	Sufficiency Clause: All deliverables supplied with the equipment shall be sufficient for satisfactory operation of the system. If any additional item is thereafter required for satisfactory operation of the system as described in the tender, the same shall be supplied by the firm with no cost implication.
11.4	The cable laying shall be carried out by HSL and the connectorization, STW, HATs and SATs are in Supplier's scope.
11.5	Preservation, Packaging and Shipping: The stores (including OBS and B&D Spares) shall be supplied in long-term preserved condition that is suitable for storage under tropical high humidity conditions for a period of 24 months.


DGM(ED)


M(ED)


CE(ED)
 (R - RINU)

Scope of Supply for SPT System per ship as below:

S.No.	Description of Electrical / Equipment with accessories	Item code	Qty./ Ship
1	SPT System consisting of		01 Set
1.1	24 Way SPT Bulkhead Mount	4524BH	23 Nos
1.2	12 Way SPT Bulkhead mount	4512BH	28 Nos
1.3	Audio/Visual (Flasher) Unit with Relay	456007	8 Nos
1.4	Headset with Microphone & 10m Cable	457001	7 Nos
1.5	12 Way Junction Box	454012	5 Nos
1.6	24 Way Junction Box	454024	2 Nos
1.7	12 Way SPT with headset	4512BS	14 Nos
1.8	24 Way SPT with headset	4524BS	9 Nos
1.9	Weatherproof Enclosure	459001	7 Nos
1.10	Telephone directory shall be manufactured and supplied only on receipt of detailed list of address from HSL/Ship		1 Set
2	On Board spares 02 years operation shall be as follows:		01 Set
2.1	Headset with Microphone	457001	01 No
2.2	Magneto Generator for SPT	457006	01 No
2.3	Selector Switch 12 Way	457004	01 No
2.4	Selector Switch 24 Way	457026	01 No
2.5	Knob for Selector Switch	457007	01 No
2.6	Magneto Generator handle for SPT	457003	01 No
2.7	12 Way SPT PCB EL 4087	457011	01 No
3	Drawings & Documentation		09 Sets
3.1	Soft copy in CDs		02 Nos